

Investigating the Effects of Sweating Avatars on Physiological and Perceptual Responses During Low- and High-Intensity Cycling in Virtual Reality

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Figure 1: User (left) wearing a VR headset, a VO₂ mask, a heart rate sensor attached on the left forearm, and the Vive tracker attached on the left lower leg, as well as the non-sweaty and sweaty male- and female-gendered avatars (right).

Abstract

Virtual reality (VR) enables users to experience avatars as their own body. Avatars can induce the Proteus effect — a shift in behavior and perception conforming to the avatar's appearance. Previous work found that sweaty avatars can reduce the perceived exertion while cycling in VR. However, it is unknown if the effects of sweaty avatars depend on the cycling intensity and if they can also influence physiological responses. Hence, we conducted a study to investigate the effects of sweaty avatars on breathing responses and perception of effort while cycling at low and high intensities. We found that participants' oxygen consumption and

perception of effort was only influenced by the sweaty appearance of the avatar during low-intensity cycling. Results suggest that an avatar's sweaty appearance can influence breathing and perceptual responses, however, only during low-intensity workloads. Our work contributes to the Proteus effect on physiological and perceptual responses in VR.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; **Virtual reality**.

Keywords

Proteus effect, virtual reality, avatar, heart rate, sweat, cycling, breathing, perception of effort

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1 Introduction and Background

In recent years, VR has evolved into a promising technology for fitness applications and exergames [4, 10, 25]. To create an embodied experience during physical activity, designers and researchers employ avatars providing users with a body in VR [31]. When users perceive the virtual world from a first-person perspective and interact via the avatar, they can experience the avatar as their own body – a phenomenon commonly known as the body ownership illusion (BOI) [15, 32].

Researchers found that the virtual embodiment of avatars with certain visual appearances causes users to change their behavior and attitude to mimic the avatars [17, 35]. Yee and Bailenson [35], for example, revealed that participants behaved more confidently in a VR dialogue when embodying attractive compared to non-attractive avatars. The authors attributed such effects to the Proteus effect [35, 36]. While the Proteus effect has been evidenced in a variety of contexts [5, 29], this phenomenon also occurs during physical effort.

Kocur et al. [22] revealed that muscular avatars can increase grip strength and reduce the perception of effort during physically demanding tasks. In line with the Proteus effect [35], the authors explain that muscularity is associated with enhanced physical abilities and, therefore, participants adapt behavior to conform with the virtual appearance [12, 22]. Interestingly, previous work showed that avatars can even influence physiological responses to physical effort. Kocur et al. [20] found that athletic avatars can decrease heart rate while cycling in VR. The findings indicate that avatars can be leveraged to positively affect users during physical exertion and, therefore, possibly create more effective exercise applications [14, 17]. However, changing an avatar's athletic appearance using stereotypical characteristics, such as the level of muscularity and body fat, can dissociate the users from the avatar due to a discrepancy between the self and the virtual body [23]. Hence, finding other visual attributes that are associated with enhanced physical abilities without drastically changing morphological characteristics could prevent users from dissociating from the avatar.

A natural human response to physical effort is sweating. Consequently, Kocur et al. [18] utilized sweaty avatars to reduce the perception of effort while cycling in VR. The authors argue that avatars should be designed with a sweaty appearance to create a realistic and lifelike VR experience and make users perceive exercise as less intense [18, 21]. However, whether sweaty avatars are effective for positively affecting the embodying users could depend on the exercise intensity. A sweaty appearance for high intensities could be perceived as positive, for low intensities, however, users could associate the avatars with fatigue and reduced physical abilities. It is, however, unknown whether visualizing sweat depends on the intensity of the physical activity. Therefore, it is crucial to understand the effects of sweaty avatars on users' physiological and perceptual responses, as the wrong usage of avatars could be detrimental to users and their experience during exercise in VR.

In this paper, we investigate whether the effects of an avatar's sweaty appearance depend on the intensity of cycling in VR. We

conducted a study with 17 participants who embodied avatars with a non-sweaty and sweaty appearance while cycling in VR with a low or high intensity. We found that participants' oxygen consumption was influenced by the intensity and avatar's sweaty appearance. Accordingly, we found that participants had a higher perception of effort over time while embodying the sweaty avatar compared to the non-sweaty avatar during low-intensity cycling. Results suggest that whether sweaty avatars are effective to positively influence users' physiological and perceptual responses depend on the exercise intensity. Designers and researchers need to be aware that visualizing sweat on avatars can create more realistic VR experiences on the one hand, however, they can also increase perception of effort during physical activities with a low intensity on the other. Our work contributes to the utilization of avatars for VR exercise systems.

2 Method

To learn about the effects of avatars with a sweaty appearance on physiological and perceptual responses, we conducted an experiment with participants that cycled with different intensities in VR while embodying sweaty and non-sweaty avatars.

2.1 Study Design

We conducted a study in a within-participants design with the two independent variables AVATAR (*non-sweaty* vs. *sweaty*) and INTENSITY (*low* vs. *high*). Hence, participants embodied an avatar of their identified gender with either a non-sweaty or sweaty appearance while cycling with a low or high intensity. All avatar conditions had the same build and wore the same clothing. The sweaty avatars were textured to have sweat covered hair, skin, and clothes. The appearance of the avatar did not change over time. INTENSITY influenced how much power was required to pedal the bike. In accordance to the bicycle protocol by Northridge et al. [26], we determined the cycling intensity based participants' body weight to ensure consistent workload across the sample (see Table 1). To mitigate sequence effects, we counterbalanced the order of conditions using a 4 x 4 Latin square.

2.2 Measures

VO₂, heart rate, and pedaling cadence were continuously monitored to quantify participants' breathing and physical responses. Perceived exertion was assessed three times during each exercise using the Rating of Perceived Exertion (RPE) scale [7]. After each condition, participants completed the Self-Perceived Fitness (SPF) questionnaire [8], the Body Representation Questionnaire (BRQ) [3, 18, 20] to assess body ownership of the avatars, and the Player Identification Scale (PIS) [13] to measure embodied presence and identification with the avatars. The Igroup Presence Questionnaire (IPQ) [30] was used to evaluate the sense of presence.

2.2.1 VO₂. VO₂ is an indicator of cardiovascular fitness and endurance performance and was measured during each condition using the VO₂ Master mask¹ – a mask worn over the mouth and nose. During exercise, the VO₂ mask measured the volume of oxygen consumed via participants' breath, as well as the volume of

¹<https://vo2master.com/store/>

Weight (kg)	Low Intensity (W)	High Intensity (W)	N
55	35	65	3
60	40	70	6
65	40	80	2
70	45	85	0
75	45	90	4
80	50	95	1
85	55	100	1

Table 1: Low and high intensities in W based on participants' body weight in kg, and the number of participants cycling with the respective intensities (N). The intensities were selected based on the standardized protocol by Northridge et al. [26].

carbon dioxide exhaled. Since oxygen consumption can vary significantly depending on an individual's body weight [16], analysis was conducted using the relative measurements (ml/kg/min).

2.2.2 Perceived Exertion. During each condition, we assessed participants' perceived exertion using the Rating of Perceived Exertion (RPE) scale developed by Borg [7]. We recorded this data three times during the condition, once after 2 minutes, again after 4 minutes, and a final time when finishing the condition after 5 minutes.

The data was acquired by displaying a scale in front of the participant within the virtual scene ranging from 6 to 20, where 6 is no exertion and 20 is maximum exertion. The experimenter asked the participant to rate their level of exertion relative to the displayed scale. Participants would then verbally convey their response, which was recorded by the experimenter.

2.2.3 Heart Rate. We used an optical heart rate sensor (TICKR FIT heart rate monitor) and a mobile phone running the Wahoo mobile fitness application to continuously monitor participants' heart rates for each condition. The heart rate sensor was worn on participants' forearm for the duration of their participation and was also used before the exercises began to determine their resting heart rate.

2.2.4 Pedaling Cadence. The cadence was measured to determine the rate at which participants completed a full rotation of the pedals to see how this was influenced by the visual appearance of the avatars. The KICKR Core and associated application record this data continuously during each condition.

2.3 Participants

We recruited 17 participants for our study, 15 male and 2 female. They ranged from ages 21 to 30 ($M = 24.8$, $SD = 2.18$). The participants' heights ranged from 170 to 195 cm ($M = 177.0$, $SD = 7.2$). Body weights had a mean value of 68.54 kg ($SD = 8.85$). Resting heart rates ranged from 70 to 100 bpm ($M = 85.2$, $SD = 8.56$). Body Mass Index (BMI) ranged from 19.11 to 26.44 ($M = 26.86$, $SD = 2.37$). All participants were students, with 5 of them additionally employed as software engineers or developers. Participants were informed that they could withdraw or discontinue participation at any time without penalty.

2.4 Apparatus

We used an HTC Vive Pro Eye to immerse participants in the virtual environment. The HTC Vive Pro Eyes features a 1440 x 1600 pixels per eye resolution OLED display using a field of view of 110 degrees and a refresh rate of 90 Hz. We attached one HTC Vive tracker on participants' left leg to track the pedaling movements and transfer them onto the avatar.

We created the VR application using the game engine Unity (v. 2019.3.13f1). It consists of a virtual fitness room with weights in the corners and a wall-length mirror. The mirror was used to enable participants to constantly see their avatar as they ride the bike. Participants experienced the virtual environment from a first-person perspective while holding the bike's handles throughout each condition.

We created the avatars using Daz3D (v. 4.12), and the roughness and diffuse maps for the sweaty textures were created using Substance Painter (v. 2019.3.3). The four avatars are adapted from the work by Kocur et al. [19]. We animated the avatars using the inverse kinematics (IK) script of the Final IK package by RootMotion².

We used a professional racing bicycle modified with the KICKR Core³ and the KICKR Climb⁴ that is used by professionals and hobby athletes [33]. The KICKR Core is a smart bike trainer produced by Wahoo Fitness, with a variety of features to customize exercise as well as record and analyze performance data.

2.5 Procedure

Participants were welcomed and asked to sign an informed consent form before starting the experiment. They then completed a digital form covering demographics, VR experience, and basic medical details, such as visual impairments or leg injuries. Next, they put on a heart rate sensor on their forearm, and their height, weight, and resting heart rate were measured. The weight was used to determine the Watt levels for high and low intensity conditions based on the STEEP Protocol [26].

The bicycle seat height was adjusted for comfort, and the VO_2 mask was calibrated if needed. The experimenter explained the procedure and instructed participants to keep their hands on the handlebars during the exercise. Once seated, participants put on the Vive motion tracker, VO_2 mask, and VR headset with assistance as needed.

The Unity scene was initiated, featuring either the sweaty or non-sweaty avatar based on the participant's identified gender. The KICKR Core was set to the required wattage, and participants cycled at a comfortable pace for 5 minutes. Wattage remained constant throughout. The Borg Scale [34] appeared at the 2- and 4-minute marks, and participants verbally reported their exertion rating, which was recorded and the scale disappeared from the scene. At 5 minutes, they stopped and provided a final exertion rating.

Participants then removed the VR headset, dismounted, and completed the BRQ, PIS, and SPF on a computer. A 5-minute rest period followed, allowing participants to remove the VO_2 mask. This process repeated for each condition. Participants were not

²Final IK by RootMotion: <https://assetstore.unity.com/packages/tools/animation/final-ik-14290>

³<https://wahoofitness.com/devices/indoor-cycling/bike-trainer-bundles/zwift-bundles/kickr-core-zwift-buy>

⁴<https://wahoofitness.com/devices/indoor-cycling/accessories/kickr-climb-buy>

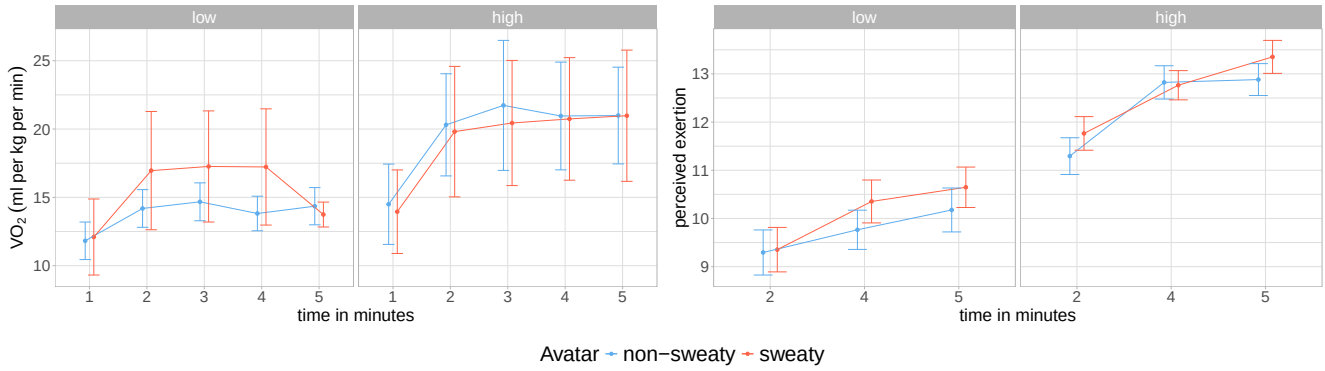


Figure 2: Mean VO₂ in ml/kg/min (left) and average perceived exertion (right). Error bars show the standard error of the mean.

informed of their performance or intensity levels. The study lasted approximately 60 minutes per participant.

3 Results

We employed a 2(AVATAR: *non-sweaty* and *sweaty*) $\times$ 2(INTENSITY: *low* and *high*) repeated-measures analysis of variance (ANOVA) on the PIS, BRQ, and SPF. To learn about the effects of the independent variables on VO₂, perception of effort, heart rate, cadence, and burned calories over time, we included the factor TIME in the statistical analysis. For VO₂ and burned calories TIME consisted of 5 levels (after 1, 2, 3, 4, and 5 minutes). For perceived exertion, TIME consisted of 3 levels (after 2, 4, and 5 minutes) as we surveyed the perception of effort three times. We analyzed the time course of the heart rate and cadence per 30 seconds, therefore, TIME consisted of 10 levels (after 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 minutes). We used R (version 4.2.2, [28]) for all statistical analyses.

3.1 VO₂

Results did not show a significant effect of AVATAR, $F(1, 13) = 0.007$, $p = .937$, $\eta_p^2 < .001$, on the VO₂ over time. However, there was a significant effect of INTENSITY, $F(1, 13) = 87.855$, $p < .001$, $\eta_p^2 = .871$, and of TIME, $F(1.77, 22.98) = 63.702$, $p < .001$, $\eta_p^2 = .831$. There was no significant interaction effect of AVATAR $\times$ INTENSITY, $F(1, 13) = 2.747$, $p = .121$, $\eta_p^2 = .174$, and of AVATAR $\times$ TIME, $F(4, 52) = 2.323$, $p = .069$, $\eta_p^2 = .152$. However, we found a significant interaction effect of INTENSITY $\times$ TIME, $F(2.21, 28.68) = 7.924$, $p = .001$, $\eta_p^2 = .379$, and of AVATAR $\times$ INTENSITY $\times$ TIME, $F(4, 52) = 2.625$, $p = .045$, $\eta_p^2 = .168$.

Due to the significant three-way interaction effect of AVATAR $\times$ INTENSITY $\times$ TIME, we also performed a Bonferroni-corrected post-hoc repeated-measures ANOVA using AVATAR and INTENSITY as factors at every time point. Our results reveal significant differences between INTENSITY at all time points t1 ($p = .030$), t2 ($p = .009$), t3 ($p = .028$), t4 ($p = .050$), and t5 ($p < .001$). We also found a significant interaction effect of AVATAR $\times$ INTENSITY at the last time point t5 ($p = .034$). Thus, results suggest that over time participants' oxygen consumption can be significantly affected by the intensity of cycling and also the avatar. Figure 2 shows the average VO₂ per condition over time.

3.2 Perceived Exertion

Results did not show a significant effect of AVATAR, $F(1, 16) = 1.490$, $p = .240$, $\eta_p^2 = .085$, on the perceived exertion over time. However, there was a significant effect of INTENSITY, $F(1, 16) = 90.693$, $p < .001$, $\eta_p^2 = .850$, and of TIME, $F(2, 32) = 42.749$, $p < .001$, $\eta_p^2 = .728$. We also found a significant interaction effect of INTENSITY $\times$ TIME, $F(2, 32) = 4.316$, $p = .022$, $\eta_p^2 = .212$, and of AVATAR $\times$ INTENSITY $\times$ TIME on the perceived exertion over time, $F(2, 32) = 3.606$, $p = .039$, $\eta_p^2 = .184$. There was no significant interaction effect of AVATAR $\times$ INTENSITY, $F(1, 16) = 0.012$, $p = .913$, $\eta_p^2 < .001$, and of AVATAR $\times$ TIME, $F(2, 32) = 0.852$, $p = .436$, $\eta_p^2 = .051$. Due to the significant three-way interaction effect of AVATAR $\times$ INTENSITY $\times$ TIME, we also performed a Bonferroni-corrected post-hoc repeated-measures ANOVA using AVATAR and INTENSITY as factors at every time point. Our results reveal significant differences between INTENSITY at all time points t1 ($p < .001$), t2 ($p < .001$), and t3 ($p < .001$). Fig. 2 shows the perceived exertion across time.

3.3 Heart Rate

Results did not find a significant effect of AVATAR, $F(1, 15) = 2.428$, $p = .140$, $\eta_p^2 = .139$, on heart rate responses. However, there was a significant effect of INTENSITY, $F(1, 15) = 24.299$, $p < .001$, $\eta_p^2 = .618$, and of TIME, $F(2.19, 32.92) = 11.605$, $p < .001$, $\eta_p^2 = .436$. There were no significant interaction effects (all $p > .05$). While descriptive statistics suggest that the heart rate is higher for the non-sweaty avatar compared to the sweaty avatar over time (see Fig. 3), we could not find any significance of the avatar's sweaty appearance. We only found effects of intensity and minute suggesting an increased heart rate for higher intensities with an increase over time (see Fig. 3).

3.4 Cadence

Results did not show a significant effect of AVATAR, $F(1, 14) = 0.019$, $p = .892$, $\eta_p^2 = .001$, and of INTENSITY, $F(1, 14) = 0.299$, $p = .593$, $\eta_p^2 = .021$ on participants' cadence. There was a significant effect of TIME, $F(3.74, 52.31) = 13.734$, $p < .001$, $\eta_p^2 = .495$. We also found a significant interaction effect of AVATAR $\times$ INTENSITY, $F(1, 14) = 5.902$, $p = .029$, $\eta_p^2 = .297$. All other interactions effects were not significant (all $p > .05$). Hence, participants had a higher pedaling

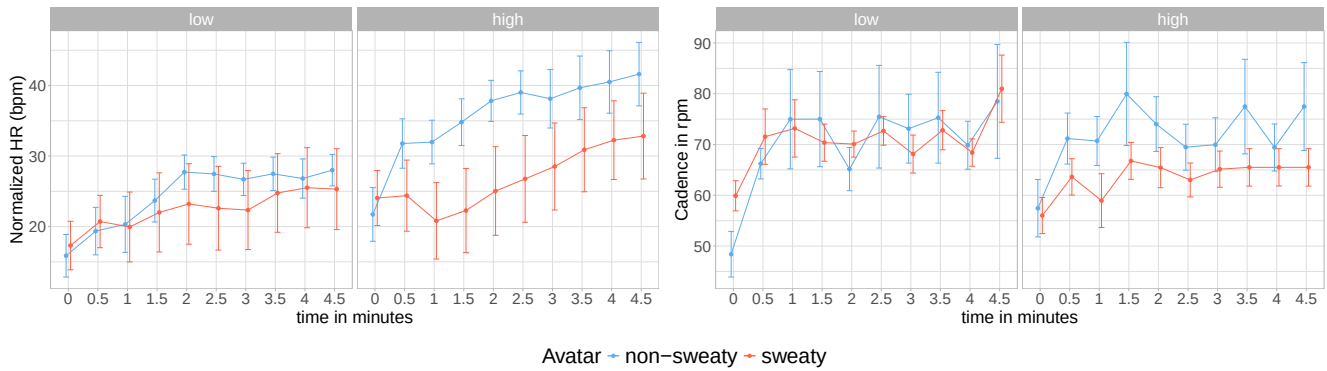


Figure 3: Average normalized heart rate relative to the resting heart rate in bpm (left) and average cadence in revolutions per minute (rpm) (right). Error bars show the standard error of the mean.

frequency while embodying the non-sweaty avatar compared to the sweaty avatar during high-intensity cycling (see Fig. 3).

3.5 Presence, Body Ownership, Avatar Identification, and Self-Perceived Fitness

We performed 2×2 ANOVAs on each dimension of the IPQ, BRQ, PIS, and SPF, and did not find any significant effects of the independent variables (all $p > .05$). Hence, this indicates that participants' experienced presence, body ownership, identification with the avatar, as well as self-perceived fitness were not different when embodying the avatars during low- and high-intensity cycling.

4 Discussion

Results suggest that participants had a higher oxygen consumption during low-intensity cycling over time while embodying the sweaty avatar compared to the non-sweaty avatar. Similarly, we also found that the perception of effort was higher for the sweaty avatar over time during low-intensity cycling compared to the non-sweaty avatar. As we achieved similar ratings for experienced embodiment and identification and could not find any significant differences between the avatars, we rule out effects caused by the sense of embodiment and identification with the avatar [9]. Overall, findings suggest that the effects of sweaty avatars depend on the exercise intensity.

4.1 Breathing and Perceptual Responses

Results show that embodying a sweaty avatar increases oxygen consumption, as indicated by elevated VO_2 values, but only during low-intensity cycling. This effect aligns with an increased perception of effort, which was higher for the sweaty avatar compared to the non-sweaty version. Given the established link between perceived exertion and VO_2 [2, 11], we suggest that a heightened perception of effort influences physical exertion, thereby increasing VO_2 .

During low-intensity cycling, the sweaty avatar likely signaled fatigue and high exertion, priming participants to perceive the task as more demanding and exert more effort, resulting in increased oxygen demand. Despite a constant workload across conditions, behavioral adaptations influenced by the avatar's appearance may

explain the physiological changes. Since the VO_2 effect diminished in the final minute (see Fig. 2), further research is needed to better understand the long-term impact of sweaty avatars on breathing responses.

According to the Proteus effect, the sweaty avatar might have influenced body posture or pedaling style, altering cycling intensity and VO_2 . Sports physiology research indicates that cycling technique, including body position and pedaling dynamics, affects power output [1]. Participants may have unconsciously adopted a more dynamic posture. While effects on cadence were observed only in high-intensity cycling, future research should examine posture changes to explain physiological variations caused by avatar appearance.

The VO_2 effect was observed only in low-intensity cycling (see Fig. 2), suggesting that avatars are more influential at lower workloads when psychological factors have a greater impact, e.g., motivation and self-perception [6, 24]. At higher intensities, increased physical demands may reduce focus on the avatar, limiting its effect. Future studies should explore higher intensity conditions to confirm this hypothesis.

No significant effect of avatar appearance on heart rate was found. However, descriptive data suggests heart rate was higher for the sweaty avatar during high-intensity cycling. This trend aligns with Kocur et al. [20], who found athletic avatars lowered heart rate. A higher cadence with the non-sweaty avatar might have contributed to an elevated heart rate (see Fig. 3). Participants may have perceived the non-sweaty avatar as less fatigued, increasing their effort through higher pedaling frequency. Future studies with larger and more diverse samples should further investigate the relationship between sweaty avatars and heart rate.

4.2 Limitations and Future Work

The main limitation is the small sample size, which may limit statistical power. To address this, we used a repeated-measures design to control for individual differences and adjusted task difficulty based on body weight. Following Kocur et al. [18], we limited conditions to two avatar types (sweaty and non-sweaty) to minimize variability. Nonetheless, statistical power may still have been insufficient to detect effects on heart rate. Additionally, the sample was

skewed toward men (15 males, 2 females), highlighting the need for future studies with larger and more diverse samples. We used 5-minute conditions to ensure a consistent workload within the aerobic threshold of 75% [27], crucial for minimizing fatigue effects in a repeated-measures design. However, this duration limits insights into longer-term effects, such as changes in the Proteus effect over time. Another limitation is the combined use of the VR headset and VO₂ mask, which were not designed to be worn together. This may have caused discomfort and added weight to the participant's head and neck, potentially increasing exertion and affecting the VR experience. Although counterbalancing helped mitigate these issues, future research should explore more comfortable methods for measuring VO₂ in VR applications.

5 Conclusion

This paper investigates how an avatar's sweaty appearance affects physiological and perceptual responses in VR cycling and whether these effects depend on exercise intensity. 17 participants cycled in VR with sweaty and non-sweaty avatars at low and high intensities. Results showed that the sweaty avatar increased oxygen consumption and perceived effort, however, only during low-intensity cycling. This suggests that the effectiveness of sweaty avatars in influencing user responses depends on exercise intensity. Designers and researchers need to be aware that visualizing sweat on avatars can create realistic VR experiences on the one hand, however, they can also elevate users' perception of effort and oxygen consumption during low intensities on the other. Further research with larger, more diverse samples and repeated exposures is needed to better understand these effects.

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